N SUDHA SUSHMA

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Github LinkedIn

Education

B-Tech in Computer Science — Aug 2019 - Jun 2023
Maharaja Surajmal Institute Of Technology. CGPA: 8.93/10

All India Senior School Certificate Examination (CBSE) — March 2018
Holy Child Auxilium, New Delhi. Percentage: 92 %

All India Secondary School Examination (CBSE) — March 2016
Holy Child Auxilium, New Delhi. CGPA: 9.5/10

Experience

Zendot Technologies Pvt. Ltd.
Software Developer — *Mar 2024 - Present*

- Developed scalable **GenAI** applications using **OpenAI** and **Anthropic Claude** models, integrating advanced prompt engineering techniques to **reduce response latency from 40s to 3s.**
- Built multi-agent systems using **LangGraph** and **LangChain**, transitioning from supervisor to swarm architectures with **few-shot LLM prompting.**
- Integrated enterprise tools (JIRA, GitHub, Google Meet) via **Model Context Protocol (MCP)**, ensuring seamless deployment within existing enterprise software workflows.
- Designed and implemented a **Retrieval-Augmented Generation (RAG)** pipeline using custom knowledge bases, enhancing NLP model contextual understanding.
- Worked with LangChain, LangGraph, LangFuse to monitor and optimize LLM performance and interaction patterns.
- Led **prompt optimization** initiatives to improve accuracy and relevance of AI-generated responses.

Indian Space Research Organization (ISRO)
Machine Learning Intern — *Feb 2022 - May 2022*

- Developed multivariate time-series models to predict regional rainfall using TensorFlow
- Worked with IMD gridded dataset of India
- Researched state, district, and taluka level information of natural disasters

Defence Research and Development Organisation (DRDO)
Machine Learning Trainee — *Jul 2021 - Sept 2021*

- Developed a Multi-Layer Perceptron based neural network
- Predicted key in Advanced Encryption Standard (AES) using Side Channel Analysis
- Used power traces of machines used for encryption as training data
- Worked closely with the researchers of deep learning at DRDO

Projects

- **Machine Translator (English - Hindi) (NLP, ML, DL)**
  - Built a Neural Machine Translation model to translate english sentences to Hindi. Used the hindi-visual-genome dataset for training. Used Indic-NLP library for hindi data. Achieved a BLUe score of 0.72 on n gram
- **Text Summarizer (NLP)**
  - Built an Extractive Text Summarization Model using the TextRank Algorithm. Trained on a local text file. Libraries Used: Numpy, matplotlib, networkx, nltk, sklearn, PyPDF2
- **CF based Recommendation System (NLP, DL, ML)**
  - Collaborative Filtering Based Recommender Systems using Low Rank Matrix Factorization(User Movie Embeddings) Neural Network in Keras. Dataset used is Movielens 100k uploaded on Kaggle. Achieved a validation loss of 0.78
- **Title Generation for Kaggle Kernels (ML, DL)**
  - Used Meta Kaggle Dataset to train a model to train new kernel titles for kaggle. Used Google's Word2vec embeddings and built a Sequential Model to achieve a loss of 1.04

Publications

- **Effect of GloVe, Word2Vec and FastText embeddings on English and Hindi Neural Machine Translation Systems**
  ICDAM 2022 (3RD International Conference on Data Analytics and Management) *Paper Link*

Skills

- **Languages & Frameworks:** Python, SQL, JavaScript, Flask, FastAPI
- **LLMs & LLMOps:** OpenAI GPT, Anthropic Claude, Hugging Face, GPT-3/4, LangChain, LangGraph
- **Cloud & MLOps:** AWS Bedrock
- **Data & Datastores:** PostgreSQL
- **Other** Machine Learning, Deep Learning, Natural Language Processing, Neural Machine Translation, RAG, RNN, LSTM, GRU, Attention Models, Transformers, Text Summarization